
AR 513: Unmanned Aerial Systems (UAS)

Course Code	AR 513
Course Name	Unmanned Aerial Systems (UAS)
L-T-P-C	3-0-0-3
Prerequisites	Consent of faculty advisor
Intended for	UG, PG and PhD
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Introduction:** Historical background of UAS, Current trends in UAS, Introduction to Unmanned Aerial Robotics (UAVs) and quadrotors. (5 hours)
- **Geometry and Mechanics:** Frame Rotations, Representations and Coordinate Systems, Kinematics and dynamics of system model, Derivation of Aerodynamic Forces. (6 hours)
- **Perception and State Estimation:** Sensors on-board, Inertial sensing, Concepts of Kalman Filtering, Inertial Navigation System design. (8 hours)
- **Flight Control:** Planar and three-dimensional dynamic models, Linear controllers for these models, Proportional Integral Derivative control, Linear Quadratic Regulator control, Linear Model Predictive Control. (8 hours)
- **Path Planning:** Collision-free Navigation, Structural Inspection Path Planning, Target Follow, Coordinated Motion, Collaborative Aerial Manipulation, Autonomous Exploration. (7 hours)
- **Final project:** Student project towards simulation and design of UAS. (8 hours)

Text Books

1. Paul Gerin Fahlstrom, Thomas James Gleason, **Introduction to UAV Systems**, Wiley.
2. Reg Austin, **Unmanned Aircraft Systems: UAVS Design, Development and Deployment**, Wiley.
3. R. Kurt Barnhart, Douglas M. Marshall, Eric Shappee, **Introduction to Unmanned Aircraft Systems**, CRC Press.

References

1. Kenzo Nonami et. al., **Autonomous Flying Robots: Unmanned Aerial Vehicles and Micro Aerial Vehicles**, Springer.
2. Kimon P. Valavanis, George J. Vachtsevanos, **Handbook of Unmanned Aerial Vehicles**, Springer.